

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
		(iii)	Concept: $E_{\text{total}} = E_k + E_p$ (1) Subs: $E_{\text{total}} = \frac{1}{2} \times 0.1 \times (3)^2 + 0.1 \times 9.81 \times 0.22$ (1) ecf on v_H but not incorrect v used, and h. $E_{\text{total}} = 0.67$ [J] (1) Alternative Total E at h = initial KE (1) can be implied $= \frac{1}{2} \times 0.1 \times 3.66^2$ (1) $= 0.67$ [J] (1)	1	1 1		3	3	
	(c)		Energy at C = $0.5 \times 0.1 \times (3.3)^2$ (1) $= 0.54$ [J] (1) Valid method for showing loss is approximately 20% e.g. 0.67 ecf $\times 0.8 = 0.54$ [J] so claim correct (1) Alternative: Ratio of KE attempted (1) Answer = 0.81 or 81% (1) Hence 19% drop so claim correct (1) Alternative: $0.8 \times 0.67 = 0.536$ [J] (1) $v = \sqrt{\frac{2 \times 0.536}{0.1}}$ (1) $v = 3.27$ [m s⁻¹] close to 3.3 so claim correct (1)			3	3	3	
			Question 3 total	6	6	3	15	11	0